

Yashvir Singh

Greater Noida, Uttar Pradesh | Phone no.: +91 9821808455 | yashvir.126@gmail.com | <https://github.com/Yashvir-13>

EDUCATION

Bennett University | *Greater Noida, U.P.* Bachelor of Technology in Computer Science
(Specialization in AI) | *Expected May 2027* * **Current CGPA:** 8.58/10.0

- Relevant Coursework:** Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Operating Systems, Database Management Systems.

TECHNICAL SKILLS

- Languages:** Python (Advanced), C++ (Data Structures), Java, SQL.
- AI/ML & Deep Learning:** CNNs, RNNs/LSTMs, Transformers, Large Language Models (LLMs), RAG (Retrieval-Augmented Generation), Computer Vision, NLP.
- Tools & Frameworks:** TensorFlow, Scikit-Learn, OpenCV, MediaPipe, Pandas, NumPy, Matplotlib, Git.
- Specialized Domains:** Neurosymbolic AI, Explainable AI (XAI), Cognitive Modeling, Optical Flow, Vector Databases (ChromaDB/Pinecone).
- Platforms:** Linux (WSL), Google Colab, Kaggle, Jupyter Notebooks

TECHNICAL PROJECTS

Fathom: Philosophical Conversational Agent | *Python, Mistral-7B, RAG, Vector DB* November 2025

- Architected a **hybrid Neurosymbolic AI agent** designed for existential reasoning and belief-challenging dialogues using **RAG pipelines**.
- Implemented a **philosophical inconsistency detection** layer using knowledge graphs to ground agent responses in historical doctrines (e.g., Schopenhauerian metaphysics).
- Developed a **deterministic persona generation** module that maintains 100% character consistency over long-context window interactions.

Cinematic Frame Interpolation via Optical Flow | *OpenCV, Python, NumPy* November 2025

- Engineered an end-to-end computer vision pipeline to increase video frame rates by synthesizing intermediate frames using **Farneback** and **Lucas-Kanade** algorithms.
- Optimized motion estimation using **warping and mixing techniques**, achieving a structural similarity index (**SSIM**) of **0.9450** and a **PSNR** of **25.96**.
- Reduced perceptual loss significantly, recorded an **LPIPS score of 0.0019**, ensuring high-fidelity cinematic motion reconstruction.

Samvedna: Real-time Sign Language Translator | *MediaPipe, OpenCV, Scikit-Learn* August 2025

- Developed a sentence-level interpreter translating Indian Sign Language into English and 10 regional languages with **96.4% accuracy**.
- Built the model using **MediaPipe** for hand/pose landmark extraction and a classification pipeline for real-time inference.
- Shortlisted** for **SIH 2024** out of 400 participating teams.

ACHIEVEMENTS & CERTIFICATIONS

- Finalist (Shortlisted), Smart India Hackathon 2024:** Shortlisted for **SIH 2024** out of 400 participating teams
- Top 5th Percentile, Adobe x GeeksForGeeks GenSolve:** Advanced to Round 2 of a high-level problem-solving competition
- Deep Learning Specialization (Andrew Ng):** Comprehensive certification covering Neural Networks, Hyperparameter Tuning, and Sequence Models.
- RAGs and Vector Databases (IBM):** Specialized training in building scalable LLM applications.
- Data Structures (UC San Diego):** Mastery of algorithmic complexity and data organization.

KEY DIFFERENTIATORS

- Research Writing:** Proficient in documenting complex AI architectures and experimental results.
- Multidisciplinary Approach:** Strong ability to bridge the gap between technical AI implementation and abstract concepts (Philosophy, Cinema).